

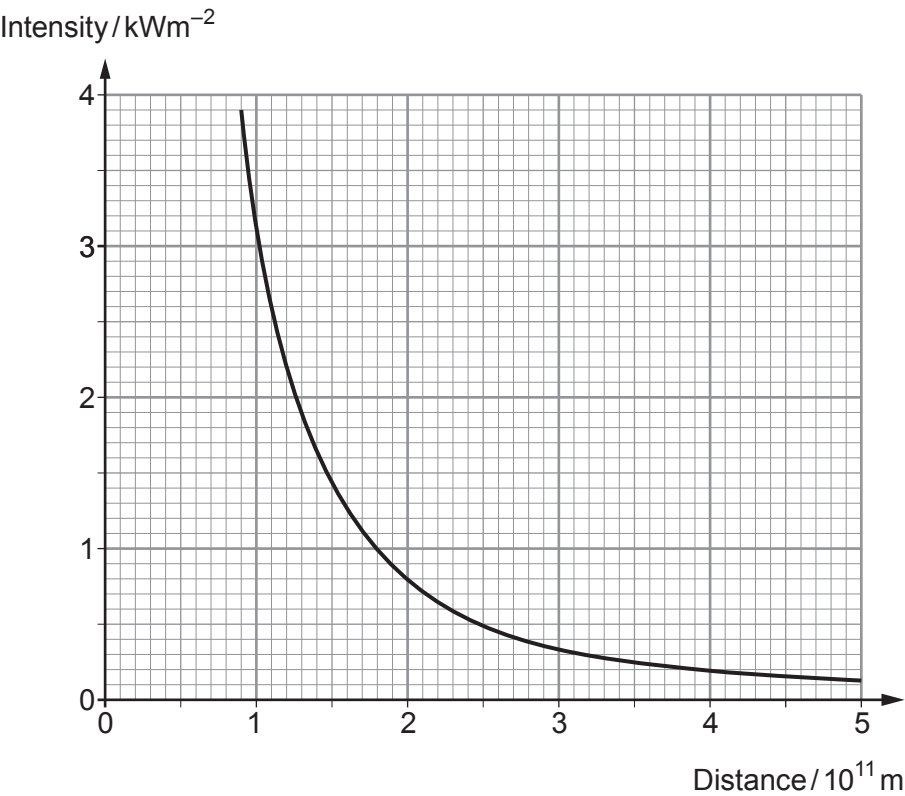
8. (a) The solar spectrum includes an absorption spectrum. Describe the appearance of this spectrum **and** state where in the Sun the absorption occurs. [2]

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- (b) The graph shows how, outside the Sun, the intensity of electromagnetic radiation from the Sun varies with the distance from its centre.



Use information from the graph to determine the total power emitted by the Sun. [3]

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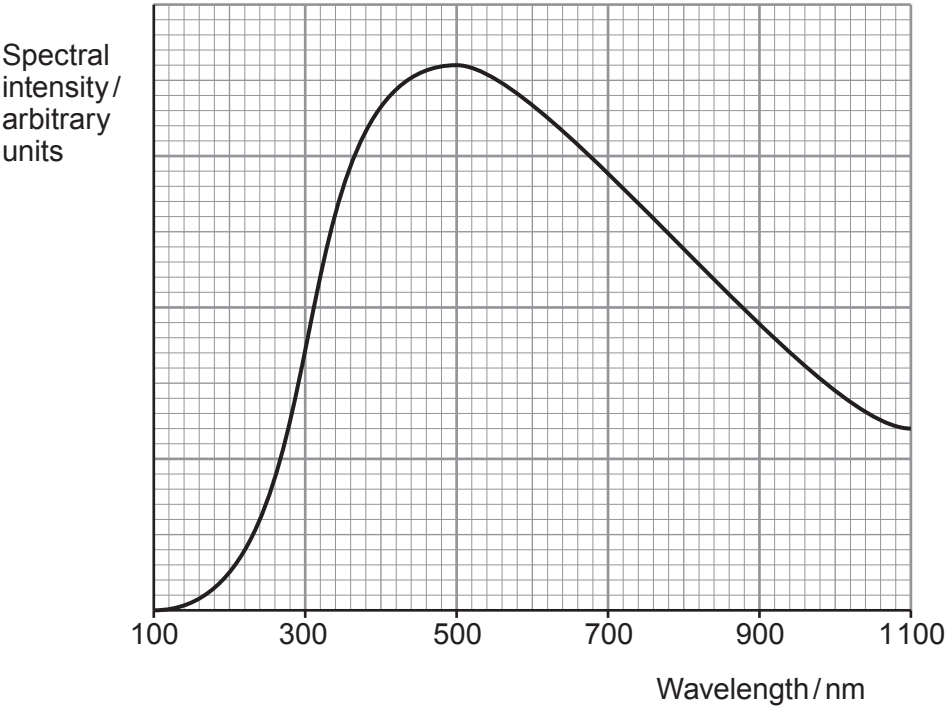
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- (c) The diagram below shows how the intensity of the radiation incident on the Earth from the Sun is distributed across the spectrum.



Determine whether or not the answer you obtained in part (b) and information which can be obtained from the above spectrum are consistent with each other.
[Surface area of the Sun = $6.2 \times 10^{18} \text{ m}^2$]

[4]

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END OF PAPER

